

QUIZ MASTER

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Summary:

It is a multi-user app (requires one admin and users, who are students) that acts as a platform for creating quizzes by the admin and attempting quizzes by users.

Materials Used:

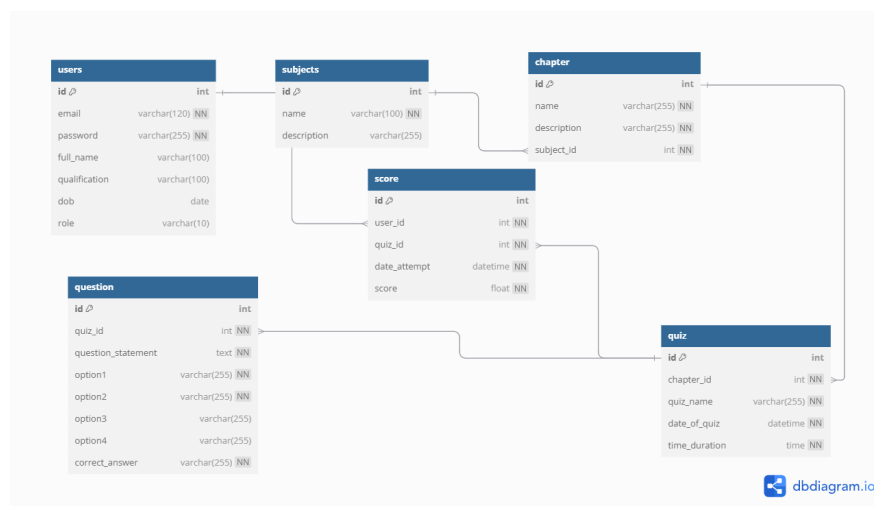
- **Frontend:** Bootstrap for styling and normal HTML.
- **Backend:** Flask, flask_sqlalchemy, chart-js, re, collections.
- **Database:** SQLite3 for storing the data of the data models.

Data Structure

The data models used are:

- **User Model:** Stores all the details of users with specific roles, whether admins or regular users.
- **Subjects:** Stores the details of the subjects and any descriptions, if needed.
- **Chapters:** Stores the details of the chapters and any descriptions, if needed.
- **Quizzes:** Stores the details of the quiz for a particular chapter, specific subject, and quiz date.
- **Questions:** Stores the details of the questions for a specific subject, chapter, and quiz. The details will be stored as the question statement, four options, and the correct answer.
- **Score:** Stores the previously attempted scores for particular quizzes by a specific user, along with the date.

Data Models



Core Functionality

- **Sign-Up and Login Functionality:**

- Separate login and sign-up forms for users, with a direct login page for the admin.
- Store details in a database named User.
- The admin can log in directly to the admin dashboard, whereas users need to sign up first and then log in to access the user dashboard.

- **Admin Dashboard:**

- **Quiz Management (Admin):**

- * Add, edit, or delete subjects and descriptions if any.
- * Add, edit, or delete chapters and descriptions of the subject.
- * Add, edit, or delete quizzes and descriptions of the chapter of that subject.
- * Add, edit, or delete questions for that particular chapter's quiz.

- **Users Details**

- * Manages all users (can view details or delete users).
- * Deletes users based on suspicion or inactivity if they are not attempting quizzes.

- **Summary**

- * You can also view a summary containing details of top scorers for quizzes, categorized by subject.

- **Search**

- * The admin can search for details about chapters and subject quizzes by their names, as well as user details by their names.

- **User Dashboard**

- **Quiz Management(User)**

- * Users can view all the quizzes based on their dates and availability.
- * Based on the availability of the quiz, the user can attempt the particular quiz.

- **Score**

- * Stores all the previously attempted quiz scores with the date of attempt, along with details such as the chapter, subject, and quiz name.

- **Summary**

- * You can also view a summary containing the number of quizzes present according to subjects.

- **Search**

- * The user can search for details about quizzes and subjects by their names.

- **Logout**

- Logout functionality is provided to both users and admins to log out from their respective sessions.

Development Approach

- **Initial Set-up**
 - Set up a **Quiz Master** folder, installed all the required libraries, and created the necessary code files for **Python**, **HTML**, **Bootstrap**.
- **Feature Development**
 - Created all the routes for HTML and buttons.
 - Created a role-based entry, i.e., for users and admins, with respective features for them.
- **Code Testing**
 - Tested all the routes for specific buttons and HTML pages.

Learnings

- **Understanding Multi-User Application Structure**
 - Developed an understanding of organizing and managing a multi-user application with distinct user roles and access levels.
- **Exploring Backend and Frontend Functionalities**
 - Gained insights into various backend and frontend features, improving the app's overall functionality.
- **Overcoming Challenges**
 - Identified and resolved multiple errors, enhancing proficiency in code syntax and practical problem-solving.

Drive Link

<https://drive.google.com/file/d/11B0CN-Q7OF-85mEAA2eFUycubl1NuSA/view?usp=sharing>